EUNGWANG SEO

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EDUCATION

University of Glasgow, United Kingdom *2022-2026*
Ph.D. in Physics and Astronomy
Thesis: Gravitational Lensing of Gravitational Waves: From Searches to Cosmology and Population Inference
Advisor: Martin Hendry and Graham Woan

The Chinese University of Hong Kong, Hong Kong *2019-2022*
M.Phil. in Physics
Thesis: Detecting and Interpreting Lensed Gravitational Waves Using Bayesian Inference
Advisor: Tjonnie Guang Feng Li and Otto Akseli Hannuksela

Kyunghee University, South Korea *2013-2019*
B.S. in Astronomy and Space Science, *summa cum laude (valedictorian)*

AWARDS AND HONORS

- Graduate School Mobility Funding Scholarship, University of Glasgow (£3k) *2024*
- Peter MacGuire Bequest (Lord Kelvin Fund), University of Glasgow (£2k) *2023*
- College of Science and Engineering Scholarship (Full funding: tuition + stipend), University of Glasgow *2022*
- President's Award (Valedictorian), Kyunghee University *2019*
- KSSS Award for honored graduates, The Korean Space Science Society *2019*
- Dean's list, Kyunghee University *2013-2018*

PRESENTATION

<Invited Presentations>

1. **Utrecht University GRASP Seminar**, *Utrecht, the Netherlands* *Oct. 2026*
Gravitational-Wave Microlensing as a Probe of Compact Object Population

2. **Chung Ang University GW Seminar**, *Seoul, Republic of Korea* *Aug. 2026*
GW231123: The Heaviest Binary Black Hole Merger — How Does Lensing Change the Interpretation?

3. **Yonsei University GW Seminar**, *Seoul, Republic of Korea* *Aug. 2026*
Probing the Hubble Constant with Strongly Lensed Standard Sirens

4. **2026 KGWG Summer Meeting**, *Po-hang, Republic of Korea* *Jul. 2026*
Measuring the Hubble Constant with Strongly Lensed Dark Sirens and Galaxy Lensing Catalogs

5. **KASI 76th Workshop on GW and NR**, *Daejeon, Republic of Korea* *May 2026*
Search for gravitational wave lensing signatures in the first part of LIGO-Virgo-KAGRA's fourth observing run

6. **KU Leuven GW Seminar**, *Leuven, Belgium* *Oct. 2025*
Strong lensing of dark sirens and galaxies as a probe of the Hubble constant

7. **UC Louvain GW Seminar**, *Louvain-la-Neuve, Belgium* *Oct. 2025*
Direct measurement of the Hubble constant from strongly lensed dark sirens via the Hubble–Lemaître law

8. **Korea Astronomy and Space Science Institute Seminar**, *Daejeon, Republic of Korea* *Apr. 2024*
Detecting and interpreting lensed gravitational waves using Bayesian inference

9. **Chinese University of Hong Kong Summer School**, *Hong Kong SAR* *Aug. 2023*
Gravitational-wave strong lensing and microlensing

10. **Korean Gravitational Wave Group monthly Colloquium**, *Seoul, Republic of Korea* *Nov. 2021*
Searching for microlensing signatures in gravitational waves

<Contributed Presentation>

1. **GWTC-4.0 CBC Lensing Webinar**, *Online* *Apr. 2026*
GWTC-4.0: Searches for Gravitational-Wave Lensing Signatures

2. **13th Belgium-Dutch Gravitational-Wave Meeting**, *Nijmegen, the Netherlands* *Oct. 2025*
Residual test to search for microlensing signatures in strongly lensed gravitational waves

3. **National Astronomy Meeting 2025**, *Durham, England* *Jul. 2025*
Detection of microlensing imprints in strongly lensed gravitational wave events using residual analysis

4. **SUPA Cormark Meeting 2023**, *Glasgow, Scotland* *Dec. 2023*
Rejection sampling for mapping model-independent parameters to lens-model-dependent properties

5. **Amaldi15**, *Online* *Jul. 2023*
Inferring properties of dark galactic halos using strongly lensed gravitational waves

<Poster Presentation>

1. **GR24 / Amaldi16**, *Glasgow, Scotland* *Jul. 2025*
Lens-model-independent analysis of strongly lensed and microlensed gravitational waves

2. **LIGO-Virgo-KAGRA Collaboration Meeting**, *Melbourne, Australia* *Mar. 2025*
Residual test to search for microlensing signatures in strongly lensed gravitational waves

3. **Theo Murphy meeting: Multi-messenger Gravitational Lensing**, *Manchester, England* *Mar. 2024*
Constraining lens model parameters of strongly lensed gravitational waves with multiple lens models

4. **LIGO-Virgo-KAGRA Collaboration Meeting**, *Toyama, Japan* *Sep. 2023*
Gravitational lensing-aided luminosity distance estimation for compact binary coalescences

5. **National Astronomy Meeting 2023**, *Cardiff, Wales* *Jul. 2023*
Reducing distance uncertainties for binary black holes by accounting for lensing

6. **LIGO-Virgo-KAGRA Collaboration Meeting**, *Chicago, IL, USA* *Mar. 2023*
Inferring properties of dark galactic halos using strongly lensed gravitational waves

7. **Asia-Pacific School and Workshop on Gravitation and Cosmology 2020**, *Daejeon, South Korea* *Feb. 2020*
Estimating dark halo mass using lensed gravitational waves

WORK EXPERIENCE AND SERVICE

Analysis & Review roles (*Analyst; Editorial team; Code and result reviewer)* *2021-present*
- LIGO-Virgo-KAGRA Collaboration and Lensing Working Group

Journal Referee *2020–present*
- Physical Review D (2020–), Physical Review Letters (2026–)

Visiting Researcher (*worked with Prof. Jose Maria Ezquiaga)* *2024*
- Niels Bohr Institute, Copenhagen

Research Assistant (*worked with Prof. Chunglee Kim and Dr. Kyungmin Kim)* *2022*
- Ewha Womans University, Seoul

Undergraduate Internship (*supervised by Prof. Hyung Mok Lee)* *2017*
- Seoul National University, Seoul

TEACHING EXPERIENCE

Graduate Teaching Assistant

1. School of Physics and Astronomy, University of Glasgow *Jan. 2023 - Jan. 2026*
- *Astronomy 1 Lab. and Astronomy 2 Lab.*

2. Department of Physics, The Chinese University of Hong Kong *Sep. 2019 - Jun. 2021*
- *Electromagnetic Theory 1, Introduction to Astronomy and Astrophysics, and Statistical Mechanics*

EXTRACURRICULAR ACTIVITIES

Student President of the Department, *Kyunghee University* *2017*

Full-time Military Service, *Republic of Korea Navy* *2014 - 2016*

TECHNICAL COMPETENCIES

Programming: `C++, Python`

Software & Tools:
GW Data Analysis: `bilby, dynesty, emcee, GWPopulation, GWpy, PyCBC`
GW & EM Lensing: `DINGO-Lensing, Golum, Gravelmaps, lenstronomy`
Cosmology: `Astropy, COLOSSUS, gwcosmo, Pyhalo`